Dietary seaweed modifies estrogen and phytoestrogen metabolism in healthy postmenopausal women.

Seaweed and soy foods are consumed daily in Japan, where breast cancer rates for postmenopausal women are significantly lower than in the West. Likely mechanisms include differences in diet, especially soy consumption, and estrogen metabolism. Fifteen healthy postmenopausal women participated in this double-blind trial of seaweed supplementation with soy challenge. Participants were randomized to 7 wk of either 5 g/d seaweed (Alaria) or placebo (maltodextrin). During wk 7, participants also consumed a daily soy protein isolate (2 mg isoflavones/kg body weight). After a 3-wk washout period, participants were crossed over to the alternate supplement schedule. There was an inverse correlation between seaweed dose (mg/kg body weight) and serum estradiol (E2) (seaweed-placebo = $y = -2.29 \times$ dose $+ 172.3$; $r = -0.70$; $P = 0.003$), [corrected] which was linear across the range of weights. Soy supplementation increased urinary daidzein, glycitein, genistein, and O-desmethylangolensin ($P = 0.0001$) and decreased matairesinol and enterolactone ($P < 0.05$). Soy and seaweed plus soy (SeaSoy) increased urinary excretion of 2-hydroxyestrogen (2-OHE) ($P = 0.0001$) and the ratio of 2-OHE:16alpha-hydroxyestrone (16alphaOHE(1)) ($P = 0.01$). For the 5 equol excretors, soy increased urinary equol excretion ($P = 0.0001$); the combination of SeaSoy further increased equol excretion by 58% ($P = 0.0001$). Equol producers also had a 315% increase in 2:16 ratio ($P = 0.001$) with SeaSoy. Seaweed favorably alters estrogen and phytoestrogen metabolism and these changes likely include modulation of colonic bacteria.